MIL-C-50446A(AR)
27 JULY 1981
SUPERSEDING
MIL-C-50446(MU)
9 July 1969

MILITARY SPECIFICATION
CASE, CARTRIDGE, 40MM, M118
PART FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of case, cartridge designated as Case, Cartridge, 40MM, M118.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-45	-	Tape, Paper, Command, Sealing and Securing
PPP-B-601	-	Boxes Wood, Cleated Plywood
PPP-B-621	-	Box Wood, Nailed and Lock Corner
PPP-T-636	-	Boxes, Fiber

MILITARY

MIL-B-2427	-	Box, Ammunition Packaging, Wood Nailed
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-46506	-	Box, Ammunition Packing, Wood Wirebound
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS (See 6.8)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8844610	-	Case, Cartridge, 40MM, M118

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-64	-	Standard Method of Salt Spray (Fog) Testing
ASTM E 18-67	-	Standard Methods of Test for Rockwell Hardness & Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Part. The part shall comply with all requirements specified on Drawing (Dwg.) 8844610 all associated drawings, and with all requirements specified in applicable specifications.

3.3 Hydrostatic pressure. The cartridge case, less vent holes, shall be capable of withstanding a hydrostatic pressure of 48,000 pounds per square inch (psi), minimum (min.), without evidence of a rupture Of the bottom of the charge cup cavity. Cases especially prepared for this test shall be identified from regular cases by painting a red stripe around the open end, or any positive identification approved by the procuring activity.

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3.4 Hardness. The hardness of the area near the mouth and on the base of the cartridge case shall be 65 ± 10 on the Rockwell 30 T Scale.

3.5 First article testing. This specification makes provisions for frist article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts shall be fabricated and finished in a thorough workmanship manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Cartridge Case	8844610	135
Cartridge Case Less Vent Holes	8844610	100

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge Case	1 OF 1	SEE BELOW
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Case (Dwg. 8844610)				
	Examination for defects	100		4.4.2.1	
	Hardness	100		4.5.2	
	Salt Spray	10		4.5.3	
	Diameter of vent holes	* 25			
	Cartridge Case, Less Vent Holes (Dwg. 8844610)				
	Hydrostatic test	100		4.5.1	

NOTES:

* Sectioned and the diameter of each vent hole shall be measured.

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4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 8844610
4.4.2.1	Case, Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Three (3) or more vent holes blocked or missing		100%	3.2	Automatic Gage (See 4.4.3.2)
MAJOR					
101.	One or two vent holes blocked or missing		100%	3.2	Automatic Gage (See 4.4.3.2)
102.	Hardness		0.40%	3.2	4.5.2
103.	Process control inspection of bath temperature for anodic coating			3.2	4.4.3.3
104.	Salt spray	*	*	3.2	4.5.3
105.	Diameter of vent hole		0.40%	3.2	Gage (See 4.4.3.2, 4.5.4)
106.	Concentricity of largest outside diameter with second largest outside diameter	See 4.4.3.	0.40%	3.2	Gage
107.	Concentricity of largest inside diameter with second largest outside diameter		0.40%	3.2	Gage
108.	Concentricity of base plug cavity with second largest outside diameter		0.40%	3.2	Gage
109.	Concentricity of powder charge cavity with base plug cavity		0.40%	3.2	Gage
110.	Diameter of powder charge cavity, min.		0.40%	3.2	Gage

NOTES:

* Sampling and rejection shall be in accordance with MIL-A-8625.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50466A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.1	Case, Cartridge				8844610
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
111.	Diameter of base plug cavity		0.40%	3.2	Gage
112.	Depth of base plug cavity		0.40%	3.2	Gage
113.	Depth from base to bottom of powder charge cup cavity		0.40%	3.2	Gage
114.	Second largest outside diameter (See 6.5)		0.40%	3.2	Gage
115.	Diameter of node		0.40%	3.2	Gage
116.	Perpendicularity of open end to second largest outside diameter		0.40%	3.2	Gage
117.	Perpendicularity of base to second largest outside diameter		0.40%	3.2	Gage
118.	Largest inside diameter (See 6.6)		0.40%	3.2	Gage
119.	Distance of flat of inside projection		0.40%	3.2	Gage
120.	Diameter of flange, max.		0.40%	3.2	Gage
121.	Total length		0.40%	3.2	Gage
122.	Diameter at datum length, min.		0.40%	3.2	Gage
123.	Unconfined second largest outside diameter		0.40%	3.2	Gage
124.	Burr in vent hole		0.40%	3.2	Visual/Equipment
125.	Metal defective		0.40%	3.6 & 6.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50466A(AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.1	Case, Cartridge		8844610
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Thickness of flange		0.65%	3.2	Gage
202.	Length of base plug seating projected from bottom of base		0.65%	3.2	Gage
203.	Wall thickness, min.		0.65%	3.2	Gage
204.	Stamping misleading or unidentifiable		0.65%	3.2	Visual
205.	Protective coating with bare spot except anoderack marks		0.65%	3.2	Visual
206.	Color incorrect		0.65%	3.2	Visual
207.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.2 Total vent hole area (alternate method). Each cartridge case shall be inspected for total vent hole area. If any cartridge case fails to comply with the requirements specified on the applicable drawing, it shall be classified defective and removed from the lot.

NOTE: If, upon calibration, the equipment setting has changed so as to permit acceptance of cartridge case having total vent hole areas that deviate from the requirement specified on the applicable drawing, then the cases accepted since the last calibration must be re-inspected and the defective parts removed from the lot.

When this method of inspection applies, the requirements of Paragraph 4.4.2.1, Critical 1 and Major 101 can be performed by a Visual/Manual method of inspection

4.4.3.3 Process control inspection of bath temperature for anodic coating. Continuous monitoring inspection shall be conducted for bath temperature requirement specified on the applicable drawings. If the bath temperature for a batch of parts does not comply with the applicable drawing requirement, the batch shall be suitably identified, classified defective and removed from the lot.

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4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Hydrostatic test of cartridge case. The cartridge case, less vent holes, shall be placed in an approved fixture. The protrusion inside of the case shall not be supported. The fixture shall be placed in a press and a load applied until the required pressure is reach. The ram shall be applied at a rate not exceeding 0.25 inch per minute. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.2 Hardness. The cartridge case shall be subjected to the Rockwell hardness test in accordance with ASTM E 18-67. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.3 Salt spray. The cartridge case shall be subjected to the salt spray test in accordance with ASTM B 117-64. This test shall be conducted for 68 hours using a salt solution of 5 percent. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.4 Total vent hole area (alternate method). The cartridge case shall be inspected for total vent hole areas, using approved equipment. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PACKAGING

5.1 Preservation and packaging. Packaging will be Level B unless otherwise specified.

5.1.1 Level A. Packaging shall be the same as Level B packaging, except that one to four Level B packs shall be overpacked in a wooden box complying with PPP-B-601, PPP-B-621, MIL-B-2427 or MIL-B-46506.

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5.1.2 Level B. The case cartridge shall be packaged for shipment as described below:

5.1.2.1 Case, cartridge, M118. Five hundred (500) cases shall be packed base down in five (5) layers in Type 1, Class 1, "B" Flute, 175 pound minimum, domestic class, RSC Style, Specification PPP-B-636 fiberboard box. Inside dimensions of fiber box shall be 13 3/4 X 18 3/4 X 9 3/4 high (± 1/16 tol). Each layer shall be composed of an egg crate type separator in a 10 X 10 pattern. Separators shall fit snug in fiber box, length and width. Height of separator to 1 1/2 inches. A filler 18 X 18 shall be used between layers and on top as required for tight packing. Material used for separators and fillers shall be the same as that used for the fiber box.

5.1.2.2 Alternate pack for case, cartridge, M118. Three hundred (300) cases shall be packed in six (6) layers in Type 1, Class 1, "B" flute, 175 pound minimum, domestic class, RSC Style, Specification PPP-B-636 fiberboard box. Inside dimensions of fiber box shall be 17 1/2 X 8 3/4 X 11 5/8 high. Each layer shall be composed of an egg crate type separator in a 10 X 5 pattern. Separators shall fit snug in fiber box, length and width. Height of separator to be 1 1/2 inches. A filler 17 X 8 shall be used between layers, and on top as required for tight packing. A spacer 8 5/8 X 8 5/8 shall be placed between flaps on bottom and on top of fiber box. Material used for separator, fillers and spacers shall be the same as that used for fiber box.

5.2 The construction of the fiber box and the material requirements of the fiber box shall comply with PPP-N-636. When assembled, the corrugations of the fiber box shall parallel the fiber box depth. The manufacturer's joint on the fiber box shall be reinforced with paper tape or metal stapled or stitched.

Spacing of staples or stitches shall not average over 1 1/2 inch apart, center to center. A manufacturing tolerance of plus or minus 1.8 inch is permissible. The parts manufacturing so that proper fit of packing material dimensions provide a proper fit of all parts, as determined to be necessary by examination of sample packs, are permissible.

5.3 Sealing. The fiber box shall be secured with gummed Kraft paper not less than 3 inches wide, comply with PPP-T-45, Grade B Tape. The tape shall be centered the full length of the outer flap seams and extend over each end of the fiber box at least 3 inches. The flap edge seams shall be similarly sealed with the tape extending at least 3 inches along the corners of the fiber box. Tape shall be properly moistened and sufficiently compressed to insure tight adhesion to the fiber box.

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5.4 Marking. Marking of the fiber box shall comply with MIL-STD-1169.

5.5 Unitization. Unitization shall be as required by the procuring agency.

5.6 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use with 40MM ammunition.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, ATTN: DRDAR-QAA-S, Dover, NJ 07801.

6.5 If the requirement cannot be met in the free state, to “go” inside diameter plug gage may be inserted in the cartridge case while inspecting the “go” and “no go” second largest outside diameter. If the “go” inside diameter plug is utilized, the data cards shall be annotated accordingly.

6.6 If the requirement cannot be met in the free state, a confining fixture may be utilized while inspecting the “go” and “no go” largest inside diameter. If the confining fixture is utilized, the data cards shall be annotated accordingly.

6.7 Metal defective. Nick, crack, dent or sharp edges which might interfere with the proper assembly or end item performance of the item.

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6.8 Drawings. Drawings listed in Section 2 of this specification under the heading of US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Asrensl, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is not under cognizance of ARRADCOM.

6.9 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-50446A CASE, CARTRIDGE, 40MM, M118 Part For

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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B. RECOMMENDED WORDING CHANGE

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